

Unit 4 Assessment of Learning - Application of Derivatives

Knowledge & Understanding	Application	Thinking	Comm.
/ 11	/10	/ 5	/2

Instructions: Please show all your work for full marks. You may only use methods taught in MCV4U for full marks. Express your final answers in **exact** form, unless otherwise indicated. The use of cellphones, audio-or video-recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

KNOWLEDGE & UNDERSTANDING - [11 Marks]

1. A position function of an object is given by $s(t) = \frac{2}{3}t^3 - \frac{13}{2}t^2 + 15t + 8, t \geq 0$, where t is in seconds and $s(t)$ is in metres.

- a. Determine the time(s) when the object is at rest. [2 Marks]

$$v(t) = 2t^2 - 13t + 15$$

$$= (2t - 3)(t - 5)$$

$$v(t) = 0 \Rightarrow t = \frac{3}{2}, 5$$

$\therefore$ At $t = \frac{3}{2}$ & 5
Seconds

- b. Determine the **exact** velocity when the acceleration is zero. [2 Marks]

$$a(t) = 4t - 13 \quad v\left(\frac{13}{4}\right) = \left(2\left(\frac{13}{4}\right) - 3\right)\left(\frac{13}{4} - 5\right)$$

$$4t - 13 = 0$$

$$v\left(\frac{13}{4}\right) = \left(\frac{7}{2}\right)\left(-\frac{7}{4}\right)$$

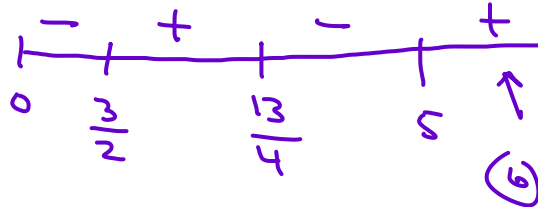
$$t = \frac{13}{4}$$

$$v\left(\frac{13}{4}\right) = -\frac{49}{8} \text{ m/s}$$

- c. Determine the interval(s) when the object is speeding up **and** the interval(s) when the object is slowing down. **Note:** use interval notation. [3 Marks]

$$v(t) \cdot a(t)$$

$$= (2t - 3)(t - 5)(4t - 13)$$



speeding up: $\left(\frac{3}{2}, \frac{13}{4}\right) \cup (5, \infty)$

slowing down: $\left[0, \frac{3}{2}\right) \cup \left(\frac{13}{4}, 5\right)$

2. Determine the **exact** value of a positive number such that the sum of 27 times the cube of the number and its reciprocal is a minimum. [4 Marks]

Let x rep. positive #

$$M = 27x^3 + \frac{1}{x}$$

$$M' = 81x^2 - \frac{1}{x^2}$$

$$M' = 0$$

$$\frac{81x^4 - 1}{x^2} = 0$$

$$81x^4 - 1 = 0$$

$$x^4 = \frac{1}{81}$$

$$x = \pm \frac{1}{3}$$

reject

$\therefore$ The # is $\frac{1}{3}$

APPLICATION - [10 Marks]

1. A farmer wants to enclose a rectangular field that has an area of 100 m^2 . The north and south sides of the field is to be enclosed with wire fencing that costs \$40 per metre and the east and west sides of the field is to be enclosed with wooden fencing that costs \$10 per metre. Determine the **exact** dimensions of the field that will minimize the cost. [5 Marks]

$$* x \cdot y = 100$$

$$C = 40(2x) + 10(2y)$$

$$C = 80x + 20\left(\frac{100}{x}\right)$$

$$C = 80x + \frac{2000}{x}$$

$$C' = 80 - \frac{2000}{x^2}$$

$$C' = 0$$

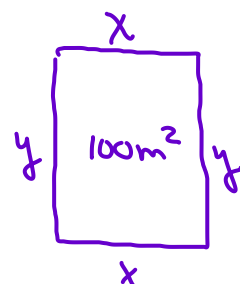
$$\frac{80x^2 - 2000}{x^2} = 0$$

$$x^2 = 25$$

$$x = \pm 5$$

reject

∴ The width is 5m and the length is 20m



2. The base of a rectangle lies along the x -axis, and the upper two vertices are on the curve defined by $f(x) = q - x^2$, where q is a constant. Determine the **exact** maximum area of the rectangle in terms of q . [5 Marks]

$$A = 2xy$$

$$A = 2x(q - x^2)$$

$$A = 2qx - 2x^3$$

$$A' = 2q - 6x^2$$

$$A' = 0$$

$$2q - 6x^2 = 0$$

$$x^2 = \frac{q}{3}$$

$$x = \pm \sqrt{\frac{q}{3}}$$

reject

$$\therefore x = \sqrt{\frac{q}{3}}$$

$$y = q - \left(\sqrt{\frac{q}{3}}\right)^2$$

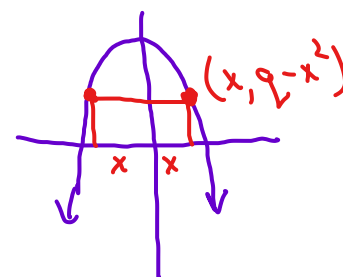
$$y = q - \frac{q}{3}$$

$$y = \frac{2q}{3}$$

$$\therefore A = 2xy$$

$$A = 2\sqrt{\frac{q}{3}} \cdot \frac{2q}{3}$$

$$A = \frac{4q\sqrt{q}}{3\sqrt{3}} \text{ units}^2$$



THINKING - [5 Marks]

1. Determine the equation (point-slope form) of the line that passes through the point (9, 15) and cuts off the least amount of area in Quadrant I that is bounded by the x -axis and the y -axis. [5 Marks]

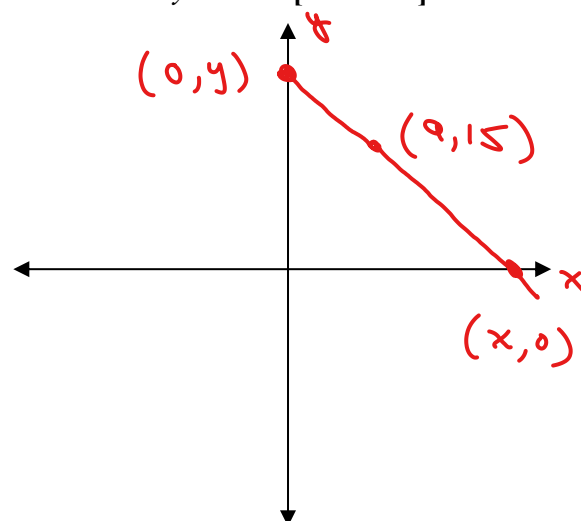
$$m = \frac{-y}{x} = \frac{y-15}{-9}$$

$$9y = xy - 15x$$

$$9y - xy = -15x$$

$$y(9-x) = -15x$$

$$y = \frac{-15x}{9-x} = \frac{15x}{x-9}$$



$$A = \frac{1}{2} b \cdot h$$

$$A = \frac{1}{2} xy$$

$$A = \frac{1}{2} x \left(\frac{15x}{x-9} \right)$$

$$A = \frac{1}{2} \left(\frac{15x^2}{x-9} \right)$$

$$A' = \frac{1}{2} \left[\frac{30x(x-9) - 15x^2}{(x-9)^2} \right]$$

$$A' = \frac{1}{2} \left[\frac{15x^2 - 270x}{(x-9)^2} \right]$$

$$A' = 0 \Rightarrow 15x^2 - 270x = 0$$

$$15x(x-18) = 0$$

$$x = 0, x = 18$$

↑
reject

$$\left. \begin{array}{l} x = 18 \\ \Rightarrow y = 30 \end{array} \right\} m = -\frac{5}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y - 30 = -\frac{5}{3}x$$

(OR)

$$y - 15 = -\frac{5}{3}(x - 9)$$

(OR)

$$y = -\frac{5}{3}(x - 18)$$